

VIGNESHWARAN RAVEENDRAN

Automotive SDV Architect | Embedded Software, ESP/BMS/ADAS & Connected Vehicle Expert

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Core Competencies

- Software-Defined Vehicles (SDV) Architecture
- Zonal & HPC E/E Vehicle Systems
- Automotive Embedded Software Development
- Electronic Stability Program (ESP) & Braking Systems
- Functional Safety (ISO 26262) Compliance
- AUTOSAR & Real-Time Operating Systems (RTOS), QNX, FreeRTOS
- ADAS & Sensor Fusion (Object Detection, Path Planning)
- Global Stakeholder & Cross-Functional Team Leadership
- Agile, Scrum & Waterfall Methodologies
- Model-Based Design (MBD) & Simulation-Driven Validation

Technical Skills

Programming & Embedded Software: C, Embedded C, Python, CAPL scripting, Hi-tech C, Real View C Compiler, MATLAB Simulink

Microcontrollers & Device Drivers: TI TDA4VH, Jetson Orin/Nano, NXP MC9S12XS128, ARM LPC2148, PIC16F877A, STM32F407, MCAL Drivers (CAN, SPI, I2C, ADC, PWM, ICAP, WDG, MCU, EEPROM)

Automotive Systems & Protocols: AUTOSAR, ESP/Braking Systems, Battery Management System (BMS), ADAS, Sensor Fusion, CAN, LIN, SPI, I2C, UART, Automotive Ethernet, SOME/IP, TSN, FlexRay

Tools & IDEs: Git, EA, Visio, Drawio, PlantUML, ETAS ASCET, Arctic Studio, Vector DaVinci, CANoe, CANalyzer, Eclipse, Visual Studio, CodeWarrior, MP LAB, GUI Guider

Testing & Validation: HIL, SIL, Unit Testing, Component Testing, Integration Testing, Test IDE, TESSY, Data Monitor Tools

Profile Summary

- Software Defined Vehicle (SDV) Architect & Automotive Technology Leader with 10+ years of experience, driving next-generation E/E, zonal, and HPC-based vehicle architectures to enable future-ready, intelligent mobility ecosystems.
- Functional Ownership across ESP (Braking) & Battery Management Systems (BMS) with demonstrated success in architecture enhancements, simulation-led validation, algorithm optimization, and reliability engineering, ensuring superior performance, safety, and ISO 26262 compliance, VES budgeting, BOM.
- Proven Engineering Leader with 10+ years of progressive experience, managing and mentoring cross-functional teams (up to 8 engineers) to led end -to-end ECE platform delivery with high-impact automotive programs across the full SDLC from requirements engineering to deployment and validation.
- Deep technical expertise across AUTOSAR, Embedded Systems, ADAS, Cloud-Edge SDV Integration, OTA, Sensor Fusion, and Microcontroller Ecosystems, with strong command of CAN, Automotive Ethernet, SOME/IP & real-time vehicle communication frameworks.
- Global collaboration experience spanning the US, Europe, China, Japan and Germany, leading stakeholder interactions [FORD, VOLVO, AUDI, JLR, GEELY, HONDA], driving Agile execution, steering architecture decisions, and consistently elevating product performance, safety, and engineering maturity.

Career Timeline



Work Experience

- Feb'25-Present | NTT Data | Systems Integration Advisor**
- Leading **Software-Defined Vehicle (SDV) initiatives**, driving E/E, Zonal, and HPC-based architectures to enable next-generation connected and autonomous vehicle platforms.
 - Designing and implementing **end-to-end SDV software solutions** with expertise in TSN, SOME/IP, DoIP, and Automotive Ethernet for high-performance vehicle networks.
 - Integrating **Cloud & Edge platforms (AWS/Azure IoT)** for OTA updates, telematics, and real-time SDV data management.
 - Conducting **functional and software architecture design**, requirements engineering, and model-based validation to ensure robust, scalable, and ISO 26262 compliant systems.
 - Decomposed** vehicle-level system architecture into domain controllers, ECUs, and software components, defining clear interfaces and communication flows across automotive networks (CAN / Automotive Ethernet).
 - Translated high-level vehicle requirements into modular architecture by creating component diagrams, interface specifications, and sequence models to support scalable Software Defined Vehicle (SDV) platforms.
 - Leading a **cross-functional team of 5 engineers**, mentoring and collaborating on simulation-driven validation, software calibration, and system performance optimization.
 - Driving **ADAS and RADAR / Camera Sensor Fusion integration**, including object detection, tracking, and path planning algorithms, to enhance vehicle intelligence and autonomy.
 - Contributed to **zonal and centralized architecture decisions**, improving safety, system efficiency, and cost-effectiveness.

Cloud & Connectivity Platforms:
AWS IoT, Azure IoT, OTA
Integration, Telematics

CI/CD: Jenkins, Git, Docker

Soft Skills

Skills



Strategic Thinking
& Decision-Making



Conflict Resolution
& Negotiation



Accountability &
Ownership



Leading &
Communication
Skills

Education

- 2015
B.E. (Electronics and Communication Engineering)
Sudharsan Engineering College, Anna University. Tamil Nadu
- 2012
Diploma (Electronics and Communication Engineering)
Seshasayee Institute of Technology. Tamil Nadu,

Feb'20-Feb'25 | Bosch | Senior Engineer

- Served as **Functional Owner for Braking Systems (IPB & ESP)**, managing a **team of 8 engineers** for global clients including Ford, Volvo, and Platform R&D.
- Led **requirements engineering and software architecture design**, ensuring robust, ISO 26262 compliant braking systems.
- Designed and developed **model-based braking software (MBD) using ETAS ASCET**, performing unit, component, integration, and validation tests.
- Optimized **braking algorithms and calibration logic**, reducing system errors and improving vehicle stability.
- Drove **cross-functional team collaboration**, mentoring engineers and coordinating with hardware, firmware, and international R&D teams.
- Contributed to **architecture decisions impacting cost, safety, and functional efficiency** across multiple platforms.
- Conducted ASPICE assessment with the cross functional stakeholders.
- Acted as a technical interface between OEM, Trie-1 and internal teams.
- Conducted RCA- Root Cause analysis of the internal and cross function team and Reduced integration defects by 30% through early risk identification.
- Git-based development workflows for software and design artefact management.
- Maintained **version-controlled architecture documentation and interface specifications**.
- Familiar with **collaborative review workflows and bidirectional traceability management**.

Apr'17-Jan'20 | Amarakosha Technologies | Software Engineer

- Developed **Battery Management System (BMS) software** for electric vehicles, integrating LTC68042 Battery Cell Controller with NXP MC9S12XS128 microcontroller to improve system reliability and efficiency.
- Designed and implemented **AUTOSAR BSW MCAL drivers** and device drivers for CAN, SPI, I2C, ADC, PWM, ICAP, WDG, MCU, and EEPROM, enhancing modularity and maintainability.
- Delivered **distributed battery monitoring and protection functionalities**, reducing master microcontroller load and improving real-time performance.
- Executed **SOC, SOH & SOP estimation algorithms**, improving accuracy & battery health management.
- Led **software validation and testing**, including unit tests, component tests, and integration tests, ensuring compliance with automotive functional safety standards (ISO 26262).
- Collaborated with **cross-functional firmware, hardware, and testing teams**, ensuring seamless system integration for EV clients.
- Streamlined **software release and configuration management**, improving traceability, version control, and defect resolution.

Jun'15-Apr'17 | Techland Automation | Firmware Engineer

- Developed **device drivers for industrial automation systems** using SPI, I2C, ADC, PWM, CCP, and UART protocols for PIC and ARM microcontrollers.
- Delivered **robust and real-time firmware solutions**, enhancing system reliability, responsiveness, and precision control for industrial clients.
- Designed **custom peripheral interfaces and microcontroller firmware modules**, reducing latency and improving device performance.
- Participated in **end-to-end firmware development**, from requirement analysis and coding to testing and deployment, ensuring timely project delivery.
- Assisted in **automation project planning and execution**, collaborating with multi-disciplinary teams to meet client-specific functional requirements.
- Implemented **fault detection and recovery mechanisms** in firmware, increasing system uptime and reliability.
- Contributed to **optimization of industrial automation processes**, improving system efficiency and reducing error rates.

Refer to the Annexure for the projects below

ANNEXURE

Model-Based Application Development for Integrated Power Brake (ESP / IPB) - BOSCH

Overview: Developed model-based application software for an **Integrated Power Brake (IPB)** system—a vacuum-free electro-hydraulic braking solution integrating the brake booster, ABS/ESP modulator, and electrical vacuum pump using **ETAS ASCET** in compliance with **ASPICE** standards.

Key Contributions & Results:

- ③ **Functional Ownership – Driver Brake Request (DBR):** Led end-to-end functional ownership, translating vehicle-level brake requirements into scalable software architecture.
- ③ **Model-Based Software Development:** Designed, implemented, and unit-tested control algorithms using ETAS ASCET, ensuring ASPICE-compliant traceability and verification.
- ③ **Brake Hardware Stiffness Learning:** Implemented adaptive stiffness learning algorithms to improve braking response and consistency across operating conditions.
- ③ **Pressure Modulation & Driving Modes:** Developed plunger-based pressure modulation logic to decouple driver input and enable configurable braking modes- **Comfort, Standard, and Dynamic**.
- ③ **System Optimization:** Contributed to reduced system complexity and improved brake pressure management, enabling seamless performance across varied driving scenarios.

Battery Management System (BMS) Development for Electric Vehicles - Amarakosha

Overview: Designed and developed a **full-featured Battery Management System (BMS)** acting as the critical interface between the battery pack and the vehicle, ensuring safe, efficient, and reliable energy management under diverse operating conditions.

Key Contributions & Results:

- ③ **Core BMS Functionalities:** Implemented Contactor Control, HV Interlock Loop (HVIL), Fuse Blow Monitoring, Leak Detection, Slow & Fast Charging control.
- ③ **Battery Intelligence Algorithms:** Developed Cell Voltage Acquisition, Cell Balancing, SOC, SOH, Power Prediction & Thermal Management algorithms.
- ③ **Safety & Protection:** Built robust monitoring and fault-handling mechanisms to trigger alarms and safely disconnect the battery pack during abnormal conditions.
- ③ **AUTOSAR MCAL Development:** Designed and implemented AUTOSAR MCAL drivers for CAN, SPI, ADC, ICAP, PWM, and GPT peripherals on **MC9S12XS128** microcontrollers.
- ③ **Battery Cell Controller Integration:** Developed device drivers for LTC6804-2 Battery Cell Controllers (BCC) and interfaced them with the BMS controller.
- ③ **Distributed ECU Architecture:** Architected and implemented battery measurement and monitoring software across multiple slave ECUs, reducing computational load on the BMS master ECU.

Proof of Concept (POC) – Object Detection & Collision Warning System (ADAS) – NTT Data

Overview: Developed an **ADAS collision warning system** using simulated traffic environments and real-time camera input to detect objects and alert drivers of potential collision risks.

Key Contributions & Results:

- ③ **Algorithm Design & Deployment:** Designed and implemented collision warning algorithms and executed them on a development board.
- ③ **Computer Vision Models:** Integrated object detection models (e.g., YOLO, SSD, MobileNet) **using** OpenCV / TensorFlow to identify vehicles, pedestrians, cyclists, and traffic signs.
- ③ **Traffic Simulation:** Enabled playback of real and simulated traffic patterns to test system behavior under multiple driving scenarios.
- ③ **Threat Assessment Logic:** Implemented time-to-collision–based risk assessment using object classification and proximity estimation.
- ③ **Driver Alert System:** Developed multimodal driver alerts, including visual warnings via GUI and audio alerts via buzzer, to enhance situational awareness.

Proof of Concept (POC) – BMS Data Visualization on HMI LCD Display

Overview: Designed and implemented an HMI-based visualization system to enable real-time communication between the Battery Management System and the driver, enhancing transparency and usability in electric vehicles.

Key Contributions & Results:

- ③ **GUI Design & Implementation:** Designed and developed an intuitive HMI GUI and deployed it on an LCD development board.
- ③ **BMS–HMI Integration:** Enabled real-time display of battery health and performance metrics through reliable BMS-to-HMI communication.
- ③ **BMS Functions Visualized:** SOC, SOH, cell voltage/current, thermal status, protection states (over-voltage, over-current, short-circuit), and fault logging.
- ③ **Driver-Centric Visualization:** Delivered user-friendly dashboards displaying range, alerts, and system warnings, supporting informed driving and energy management decisions.